Shadi Mirmohammadi

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SUMMARY

Ph.D. student in Electrical and Computer Engineering (expected May 2028) working on optically tunable THz metasurfaces: Ansys HFSS design, cleanroom fabrication of ITO devices on Fe:β-Ga2O3, and optical and THz measurement. M.S. in chemistry and six peer-reviewed papers, two as co-first author, including Nano Letters 2026.

EDUCATION

Ph.D., Electrical and Computer Engineering, University of Utah — Expected May 2028
M.S., Chemistry, University of Utah — 2025
M.Sc., Chemistry (Nanochemistry), Iran University of Science and Technology — 2020
B.Sc., Applied Chemistry, K. N. Toosi University of Technology — 2017
Graduate coursework: Advanced Electromagnetic Fields (HFSS project, Spring 2026).

TECHNICAL SKILLS

EM simulation: Ansys HFSS: Floquet ports, periodic boundaries, anisotropic and frequency-dependent materials, impedance boundaries, parametric sweeps, S-parameters

Cleanroom fabrication: maskless laser lithography (Heidelberg DWL 66+), LOR/S1813 bilayer lift-off, HMDS, O2 plasma descum, DC magnetron sputtering, thermal evaporation, rapid thermal annealing (AccuThermo AW 610), substrate polishing, Cl2/Ar plasma etch trials (ICP, PlasmaTherm SLR), wet processing (TMAH develop, resist strip)

Metrology: stylus profilometry (KLA-Tencor P-20H), four-point probe, spectroscopic ellipsometry (J.A. Woollam VASE), Keyence digital microscopy, lab SAXS

Optical and THz measurement: CW THz transmission (0.2 to 1 THz), deep-UV photoluminescence and transmission with an LN2 cryostat, nanosecond transient absorption, UV-Vis, FTIR, laser alignment

RESEARCH EXPERIENCE

Graduate Research Assistant, Electrical and Computer Engineering, University of Utah — Jan 2026 – Present

- Designed an optically switchable THz metasurface absorber on Fe-doped β-Ga2O3 in Ansys HFSS: 95.8% absorption in the dark and 82% absorption modulation at 300 GHz across a 0 to 25 S/m photoconductivity sweep (first-author paper and poster, IRMMW-THz 2026).
- Redesigned the absorber with transparent ITO resonators and ground plane so the 808 nm pump reaches the crystal through the pattern; simulations show four TE/TM channels between 238 and 682 GHz that switch by at least 10 dB at 4 to 5 dB insertion loss.
- Fabricated ITO H-shaped resonator arrays with 15 µm lines on glass and Fe:β-Ga2O3 by maskless laser lithography, bilayer lift-off and DC sputtering; the latest Ga2O3 array came through lift-off with every cell intact, and its THz transmission matched HFSS to 0.04 rms from 300 to 650 GHz using the film's DC conductivity and the measured dimensions.
- Traced incomplete lift-off on Ga2O3 to an LOR undercut that was too small, doubled the develop time to reach a 2.5 to 2.7 µm undercut and added an O2 descum, which improved feature definition on the next device.
- Sputtered and annealed ITO films on the Denton Discovery 18: a 384 nm film went from 19 to 4.9 Ω/sq after a 500 °C anneal, and four-point-probe sheet resistance was compared with values extracted from THz transmission.
- Measured THz transmission from 0.2 to 1 THz on a CW photomixing system; under an 808 nm pump, Fe:β-Ga2O3 cut the transmitted THz signal by 86% at 463 GHz and recovered to 90% of the pump-off level within 37 s of turn-off, with photoconductivity reaching 6.8 S/m at 20 W.

Graduate Researcher, Chemistry (Tang and Luo groups), University of Utah — 2023 – Dec 2025

- Measured triplet energy transfer in CdSe nanoplatelet-anthracene hybrids by nanosecond transient absorption and analyzed the kinetics in Igor Pro.
- Contributed to a time-resolved EPR study of triplet energy transfer in colloidal Si quantum dots, with Argonne National Laboratory (co-author, Nano Letters 2026).

PUBLICATIONS AND PRESENTATIONS

- Mirmohammadi, S. S., et al. "Optically Tunable Terahertz Metasurface Absorber Based on Fe-Doped β-Ga2O3." IRMMW-THz 2026, Salt Lake City, Oct 2026 (first author; poster).
- Teferi, M. Y., Somasundaran, S. M., Mirmohammadi, S. S., et al. "Spin-Polarized Anthracene Triplet States Sensitized by Si QDs Revealed by Time-Resolved EPR Spectroscopy." Nano Letters (2026).
- Mirmohammadi, S. S., et al. "Excitation-Selective Eu3+ Luminescence and Defect Signatures in GdScO3 Single-Crystal Substrates." MRS Fall Meeting 2026, Boston (poster); journal manuscript in preparation.
- Six peer-reviewed journal papers, two as co-first author (Nano Letters, Small, Nanoscale Advances, RSC Advances, Front. Chem. Sci. Eng.; 337 citations); manuscripts on GeO2 deep-UV photodetectors (equal contribution) and 10 kV β-Ga2O3 switches.

HONORS

- Best Poster Award, Excited State Processes 2025; EDGES Graduate Fellowship, College of Science, University of Utah (2023).